Algorithm Analysis and Correctness: Exercises

Use 1-based indices, $\lg = \log_2$, and $\ln = \log_e$. Unless stated otherwise, records and indices occupy a constant number of words, and basic reads, writes, and comparisons have constant cost. Show reasons and intermediate steps. Finite numerical checks do not replace general proofs.

A01-E01 What does sorting promise?

The input is $[3a, 1b, 3c, 1d]$. Letters identify records; sort by numerical key only. For each output, check sortedness, preservation of all records, and stability: (1) $[1b, 1d, 3a, 3c]$; (2) $[1d, 1b, 3a, 3c]$; (3) $[1b, 3a, 3c]$. State the valid output for an empty input.

A01-E02 Prove first-match search

Apply FIRST-MATCH to $A = [6, 2, 6, 5]$, searching separately for 6, 5, and 9. List compared indices, the returned value, and the comparison count. State an invariant and prove that the returned match is the first one. Prove termination and handle an empty input.

A01-E03 What does partial correctness leave unproved?

Program P executes `while true: skip` on every input. Program Q changes the condition of FIRST-MATCH to `i < n`, leaving everything else unchanged. Assess P's partial and total correctness. Give a smallest counterexample for Q and identify whether correctness or termination fails.

A01-E04 Count insertion sort independently

Sort $[4a, 2b, 4c, 1d]$ using the strict-comparison version in the chapter. List each iteration's comparisons and resulting array. Count shifts, key comparisons, complete inner guard tests, and array writes separately. List all inversions in the original input and compare their count with the number of shifts.

A01-E05 Change one comparison symbol

Replace `>` by `>=` in insertion sort. Trace $[2a, 2b]$ and assess sortedness, record preservation, stability, and termination. Does the shift count still equal the number of strict inversions? Identify the changes needed in the inner invariant.

A01-E06 Specify a distribution before averaging

A length-4 FIRST-MATCH search fails with probability $1/4$. Conditional on success, the first matching position is uniform over the four positions. List the probabilities of the five events and calculate the expected key comparisons. Explain whether $(n+1)/2$ can be used as an average without a specified distribution.

A01-E07 Prove a tight bound with explicit quantifiers

Let $f(n) = 5n^2 + 3n + 7, n \geq 1$. Give explicit constants proving $f \in \Theta(n^2)$. Then use the definition to refute $f \in O(n)$. State how the chosen input n depends on arbitrary candidate constants c, n_0.

A01-E08 little-o and the logarithm trap

Prove $9n \in o(n^2)$ and refute $9n^2 \in o(n^2)$. Compare $n \lg n$ with n. Their logarithms have ratio tending to 1; why does that not establish that the original functions have the same order?

A01-E09 Compare oscillating functions

Set $g(n) = n$. Consider (1) $f(n) = n(2 + (-1)^n)$; (2) $h(n) = n^2$ for even n, and $h(n) = 1$ for odd n. For each, decide membership in $O(g), \Omega(g), \Theta(g), o(g), \omega(g)$. Saying only that a limit fails to exist is insufficient.

A01-E10 Three loops, three different costs

Count `visit` operations. For positive integer n, derive an exact sum and a tight bound for: (1) $i = 1..n$, with $j = 1..i$; (2) i starts at 1 and doubles while at most n, with i visits per round; (3) $i = 1..n$, with visits at $i, 2i, \ldots \leq n$. Calculate all three exact totals for $n = 6$.

A01-E11 Bound the harmonic sum on both sides

Use the decreasing function $1/x$ to prove $\ln(n+1) \leq H_n \leq 1 + \ln n$, then derive $\Theta(\lg n)$. Specify the rectangle intervals and heights for each bound. Finally use $x/2 \leq \lfloor x \rfloor \leq x$ for $x \geq 1$ to prove $\sum_{i=1}^{n} \lfloor n/i \rfloor = \Theta(n \lg n)$.

A01-E12 Bound a sum without its closed form

For fixed real $p \geq 0$, prove $\sum_{i=1}^{n} i^p = \Theta(n^{p+1})$ by an upper bound using the largest term and a lower bound using only the latter half. Give fixed multiplicative constants. Why does the argument not establish the same formula for $p = -1$?

A01-E13 Tight logarithms do not imply tight original values

Without Stirling's formula, prove $\lg(n!) = \Theta(n \lg n)$ and $n! = o(n^n)$. Hint: in $n!/n^n = \prod_{i=1}^{n}(i/n)$, inspect the first $\lfloor n/2 \rfloor$ factors. Explain why the two conclusions are consistent.

A01-E14 Numerical size and machine overflow

A loop visits $i = 1, 2, 4, \ldots \leq N$. (1) Give the complete results for $N = 13$ and $N = 0$. (2) Express the number of visits in terms of the bit length b for positive N. (3) Assuming N fits an unsigned machine word, rewrite the update rule to avoid overflow in `2*i`. (4) Another loop visits every integer from 1 to N; what is its iteration count as a function of b?

A01-E15 Cumulative allocation versus peak space

A program allocates blocks of $1, 2, \ldots, n$ words in order. It processes and frees each block before allocating the next. Find cumulative allocation and peak auxiliary space. What changes if every block is retained until the end? Assume storage can be reused; count only the stated blocks, excluding allocator fragmentation.

A01-E16 Refine the analysis using input structure

For distinct-key $S = [2, 3, \ldots, n, 1]$, $n \geq 2$, identify all inversions. Derive insertion sort's total shifts, key comparisons, array writes, and time order. Does this contradict its tight quadratic worst-case bound? Generalize to a family satisfying $I(S) \leq cn$ for fixed constant c.

A01-E17 Attach error bounds to approximations

Use the chapter's inequalities to bound $(1 + 1/10)^{10}$ and check its actual value. Set $L = \sqrt{12\pi}(6/e)^6$ and use the bounds on α_6 to obtain a numerical interval for $6!$, retaining at least three decimal places. A calculator is allowed, but show the general inequalities before reporting decimals.

A01-E18 Integer division and a safe implementation

For $x = 17, a = 3, b = 2$, check the identities for successive floors and successive ceilings. Then use $x = 1, a = b = 2$ to test whether an inner ceiling followed by an outer floor still equals $\lceil x/(ab) \rceil$. For positive integers u, v, devise a ceiling-quotient formula without first computing $u + v - 1$. Prove that its result and intermediate addition do not exceed u.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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